REQUEST FOR QUOTATION (RFQ) & PURCHASE SPECIFICATION

Document Reference: JB-RFQ-2026-NV2-001E **Date:** June 19, 2026 **Purchaser:** Joe's Brother Clean Energy Division **Project:** NV2 Bionic sCO2 Turbine Prototype Deployment **Subject:** Technical Inquiry and RFQ for Precision-Manufactured sCO2 Turbine Core Assembly

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TO:

Attn: Sales Engineering / High-Precision Manufacturing Division [Vendor Name] [Vendor Address]

Dear Sir/Madam,

On behalf of **Joe's Brother**, we are pleased to submit this Request for Quotation (RFQ) for the high-precision manufacturing, assembly, and testing of our proprietary **NV2 Bionic Turbine Core Assembly** designed for a supercritical carbon dioxide (sCO2) closed-loop power cycle.

We request your technical evaluation, price quotation, and estimated delivery lead time for the components detailed below.

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1. TECHNICAL SPECIFICATIONS & MANUFACTURING PROCESS

Item 1.1: Bionic Helical Lattice Turbine Impeller

- **Design Profile:** Helical Lattice Triple Periodic Minimal Surface (TPMS) Gyroid structure.
- **Base Material:** Nickel-based superalloy **Inconel 740H** (certified aerospace grade).
- **Manufacturing Method:** Direct Metal Laser Sintering (DMLS) / Laser Powder Bed Fusion (LPBF) 3D Printing.
- **Machining Tolerances:** Precision post-machining tolerance of **±0.01 mm** on all mating surfaces and shaft interfaces.
- **Surface Finish:** High-precision polishing on gas-flow channels.

Item 1.2: Rotor Shaft & Bearing Assembly

- **Shaft Material:** High-tensile steel alloy with integrated thermal isolators.
- **Bearing Support: Active Magnetic Bearings (AMB)** with sub-millisecond control loop compatibility. Shaft runout tolerance must not exceed **0.001 mm**.

Item 1.3: Sealing Assembly

- **Seal Type:** Dry Gas Seals (DGS) in a Tandem Configuration.
- **Mating Faces:** Silicon Carbide (SiC) faces with an operational gas film spacing of **2 to 7 µm**.

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2. OPERATIONAL & ENVIRONMENTAL CRITERIA

- **Working Fluid:** 99.99% Pure Research Grade Carbon Dioxide (CO_2).
- **Nominal Operating Pressure: 25 MPa (250 bar)**.
- **Hydrostatic Test Pressure: 35 MPa (350 bar)**.
- **Design Temperature: 715°C** (Turbine Inlet).
- **Operational Rotational Speed:** Dynamic balancing certified for **15,000 to 150,000 RPM**.

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3. DOCUMENTATION & QUALITY CONTROL REQUIREMENTS

The selected vendor is required to provide the following documentation upon delivery:

1. **Material Test Report (MTR):** Certified chemical composition and mechanical properties of Inconel 740H.
2. **Metrology Report:** Coordinate Measuring Machine (CMM) inspection report verifying dimensional compliance.
3. **Non-Destructive Testing (NDT) Reports:** X-ray CT scan and Dye Penetrant testing reports verifying structural integrity and lack of micro-fractures in the 3D-printed impeller.
4. **Rotor Dynamic Balancing Certification:** Verifying dynamic balance at maximum rated operational speeds.

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4. SUBMISSION GUIDELINES

- **Quotation Validity:** Minimum 90 days from submission.
- **Non-Disclosure Agreement (NDA):** Signed mutual NDA is required prior to sharing the detailed CAD models (.STEP files).
- **Delivery Term:** FCA (Incoterms 2020) at manufacturer's facility.

Please confirm receipt of this RFQ and indicate your interest in bidding by submitting a technical feasibility feedback and a preliminary quotation.

For technical inquiries regarding this RFQ, please contact the undersigned.

Sincerely,

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